

Lab01: Local Sports Tournament & Bracket Manager (#54)

Introduction

The **Local Sports Tournament & Bracket Manager** is a tournament administration platform designed to manage local sports tournaments. The system generates seeded single and double-elimination tournament brackets based on initial team rankings, records live match scores, calculates tiebreakers, and updates and publishes league standings. The primary stakeholders identified for the system are the **Team Captain** and the **Tournament Director**.

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall automatically generate seeded single/double-elimination tournament brackets based on initial team rankings.	High	Pass: The generated bracket follows the initial team rankings and places valid teams in valid match slots. Fail: The system produces invalid match pairings.	Ensures a valid tournament structure is created automatically based on team rankings.
FR-002	Functional	The system shall allow the Tournament Director to record and update live scores for ongoing matches.	High	Pass: When the Tournament Director enters a valid score, the updated score is displayed for the corresponding match. Fail: The system accepts an invalid score or updates the wrong match.	Enables accurate recording of match results during the tournament.
FR-003	Functional	The system shall calculate tiebreakers when teams have equal standings.	High	Pass: When two or more teams have equal standings, the system calculates and displays the applicable tiebreaker result. Fail: The system leaves tied teams unresolved when a tiebreaker is required.	Ensures teams are correctly ranked when their standings are equal.
FR-004	Functional	The system shall update league standings automatically after a match score is submitted.	High	Pass: After a valid score is submitted, the affected teams' standings are recalculated and displayed. Fail: The standings remain unchanged or show incorrect results.	Keeps tournament standings accurate after every match.
FR-005	Functional	The system shall allow the Tournament Director to publish updated league standings to tournament participants.	Medium	Pass: After publishing, participants can view the latest standings. Fail: Participants cannot access the updated standings.	Allows teams and participants to view the current tournament status.
NFR-001	Non-Functional	The system shall recalculate standings, goal differentials, and tournament leaderboard updates within 200 ms of score submission.	High	Pass: Performance testing confirms that standings and leaderboard updates complete within 200 ms after score submission under the specified test load. Fail: Any update exceeds 200 ms.	Ensures standings remain responsive during live tournament matches.
NFR-002	Non-Functional	The system shall restrict score entry and tournament result updates to authorized Tournament Directors.	High	Pass: An authorized Tournament Director can enter or update scores. Fail: An unauthorized user can modify a match score or tournament result.	Prevents unauthorized changes to official tournament results.

Actors

1. Team Captain
2. Tournament Director
3. Tournament Participant

Use Cases

1. Generate Tournament Bracket
2. Record Match Score
3. Calculate Tiebreakers
4. Update League Standings
5. View League Standings

UML Use-Case Diagram

